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Intelligent recognition of students' behavior for smart learning environments

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The automatic detection of student behaviors is essential for improving smart classroom technologies and offering data-driven insights regarding student engagement. Nevertheless, existing methods encounter considerable obstacles caused by class imbalance, restricted annotations, and the slight visual resemblances among behavior categories. To overcome these constraints, we present a meta-learning framework that combines Vision Transformers with Prototypical Networks, improved by supervised contrastive learning and hard negative mining. The process starts by preprocessing and cropping the input images, utilizing YAML annotations to focus on behavior-specific areas. Every input is converted into patch embeddings and handled by Transformer encoders, producing distinctive feature representations. Class prototypes are subsequently derived from the support set, and query samples are categorized through distance-based metrics within a few-shot learning framework based on episodes. Extensive experiments were carried out on the SCB-05 dataset under 5-way few-shot settings to confirm the effectiveness of the proposed framework. The findings show that combining Vision Transformers with contrastive learning greatly enhances feature distinctiveness, whereas hard negative mining additionally boosts generalization. Under the 5-way 10-shot evaluation protocol, our method attains a total accuracy of 91.3% and a mean Average Precision, exceeding the performance of both baseline ProtoNet and Transformer variants without hard negative mining. Further analyses, such as class-specific assessments, confusion matrices, and embedding visualizations, validate the strength and clarity of the suggested model. These results set a new standard for recognizing student behavior and emphasize the promise of meta-learning frameworks for practical uses in education.

Keywords Smart classrooms, Student behavior recognition, Deep learning, Meta-learning, Vision transformers

In recent times, the incorporation of Artificial Intelligence (AI) and computer vision into educational technologies has changed how student participation and classroom actions are observed. Conventional techniques for evaluating student involvement, like direct observation and questionnaires, tend to be subjective, labor-intensive, and susceptible to bias. As smart classrooms and widespread digital learning spaces become commonplace, there is an increasing need for automated systems that can detect and assess student behaviors instantly[1]. These systems allow educators to gain insight into degrees of engagement, teamwork, and participation, thereby enhancing adaptive teaching methods and boosting educational results.

Computer vision has emerged as a fundamental solution for tackling this issue, especially due to progress in deep learning. Initial methods depended largely on manually designed features and traditional machine learning models, yet these techniques faced challenges in generalizing to various classroom environments. The emergence of Convolutional Neural Networks (CNNs) enhanced performance by autonomously learning spatial characteristics; however, CNN-based approaches struggle to grasp long-range dependencies in student behaviors, particularly in intricate and congested classroom situations[2]. Lately, Transformer-based models like the Vision Transformer (ViT) have shown impressive success in image classification tasks, surpassing CNNs on extensive benchmarks and attaining state-of-the-art results across various application areas[3,4].

Even with these improvements, considerable obstacles still exist in recognizing behavior in classrooms. Firstly, many current datasets are imbalanced, with specific behaviors (e.g., "reading" or "hand-raising") significantly overrepresented, while others (e.g., "on-stage interaction" or "blackboard writing") are comparatively limited. This disparity results in biased models that do not perform well across all categories[5]. Secondly, gathering

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extensive annotated datasets in educational environments is expensive and presents privacy issues. As a result, few-shot learning (FSL) methods have appeared as a hopeful avenue, allowing models to generalize from just a few labeled examples per category[6,7].

Meta-learning systems, like Prototypical Networks (ProtoNet), have shown their efficiency in few-shot classification by developing generalized embeddings capable of adjusting to new classes with limited guidance. Nevertheless, ProtoNet by itself is susceptible to noisy examples and variations within the same class. Recent studies have focused on combining contrastive learning with hard negative mining to help the network develop more distinctive embeddings and enhance class separability[8,9].

In this context, our research emphasizes utilizing meta-learning alongside sophisticated embedding techniques to address the distinct challenges of recognizing student behavior. The basis of our method lies in integrating Prototypical Networks (ProtoNet) with Vision Transformers (ViT). We supplement this framework with supervised contrastive loss (SupCon) and using hard negative mining. With these improvements, we expect to develop resilient, understandable, and wide-applicable models. The design also helps the models to cope with data imbalance well and perform well under few-shot learning cases.

While current methods have shown encouraging outcomes in recognizing student behavior, various constraints remain that impede their use in actual classroom settings. A significant issue is data imbalance, as existing datasets like SCB-05 are largely skewed towards certain categories, such as "reading" or "hand-raising," while less represented behaviors often experience misclassification. This discrepancy results in skewed models that do not encompass the entire range of classroom activities. A further drawback is limited generalization, especially in traditional CNN-based models, which frequently have difficulty in capturing long-range dependencies and nuanced behavioral signals. Consequently, their effectiveness declines noticeably when utilized in unfamiliar situations or different classroom environments.

Moreover, the meta-learning ProtoNet allows models to learn new categories with only a small number of examples. However, they are less effective when there is a lot of variation within one category, or negative cases are represented poorly.This presents a difficulty for behavior classifications where students show varying postures or expressions under the same label. Moreover, a significant obstacle exists due to the lack of interpretability in numerous current models. While they may indicate high overall accuracy, they frequently do not offer insights into performance by class or the separability of embeddings, which restricts their practical value for educators looking for actionable feedback.

These challenges emphasize the need to have a more flexible and robust method of understanding student actions. This method should be able to leverage sparse and imbalanced data. It must also be able to detect even the subtle behavioral patterns. Ultimately, the method should ensure transparency and fairness in the classification of behaviors. This method would not only enhance research in classroom analytics but also offer valuable assistance to teachers and educational systems in practical environments.

Although significant advancements have been made in automated student behavior recognition, the shortcomings of current methods emphasize the necessity for a stronger and more adaptable framework. To address these issues, this paper introduces a meta-learning approach. The approach involves the combination of Vision Transformers and Prototypical Networks to determine the general architecture. It is further augmented using guided contrastive learning and hard negative mining. All these factors aim to increase the performance of few-shot and enhance the general knowledge of the model.The key contributions of this paper can be outlined as follows:

(i) We present a hybrid meta-learning framework integrating Vision Transformers with distance-based classification for effective student behavior recognition.
(ii) We propose supervised contrastive loss utilizing hard negative mining to enhance class separability in few-shot contexts.
(iii) We conduct an extensive evaluation on the SCB-05 dataset, encompassing class-wise, few-shot, and interpretability assessments; and.
(iv) We achieve new state-of-the-art results on this dataset in terms of both accuracy and mean Average Precision (mAP).

In this study, several novel ideas were introduced to the field of few-shot behavior recognition in the context of intelligent learning. First, we introduce a prototype representation that is optimized by a Vision Transformer specifically designed to capture subtle classroom behavior indicators, on par with traditional object-centric datasets. Second, we adopt supervised contrastive learning with a carefully designed hard negative mining strategy to enhance inter-class discrimination in high low-shot settings. Third, we propose an automated cropping scheme using YAML that results in improved data quality and reduces background noise in the SCB-05 dataset, which is easier to use in developing the prototype. Fourth, we introduce a new few-shot learning benchmark for the SCB-05 dataset, providing a uniform evaluation framework for upcoming studies. Finaly, the entire pipeline exhibits significant resilience in facing the class imbalance and the domain-shift scenarios, emphasizing its real-world applicability in educational analytics.

Most of the previous investigations of datasets related to SCB have focused on either identification in videos or classification of activities over time, which have completely different goals, data types, and evaluation procedures than the fixed-image few-shot classification problem addressed in the current study.The existing SCB approaches typically operate on full video sequences, rely on time-based information and motion detection, and are evaluated based on detection-based measures such as mAP or F1 based on bounding-box annotations. Comparatively, our method supports cropped static images, with no time-related information, and aims to classify behaviors using small samples of training, using few-shot episodes, with accuracy on newly introduced classes as the primary metric. Due to such differences in the formulation of tasks, the modality of input, the type of annotation, and the method of evaluation, a simple numerical comparison with the ongoing research in SCB would not be relevant or just. As a result, we place our contribution in conceptual terms, emphasizing that our solution advances on the prior video-based systems by addressing a separate and practically meaningful problem, which is the case of static few-shot behavior recognition under limited data conditions.

The essential elements utilized in this research—specifically the Vision Transformer backbone, prototype-based classification, and supervised contrastive learning—are based on recognized techniques; however, the originality of this study is in the combined adaptation and reformulation of these aspects for the specific needs of fine-grained and low-data classroom behavior recognition. In contrast to traditional few-shot benchmarks, the SCB-05 dataset features nuanced posture variations, regular occlusion, and significant background interference, making prototype estimation and class separation especially difficult. Our contribution focuses on methodology and task execution instead of architecture: we propose a domain-specific episodic structure and prototype enhancement approach designed for nuanced behavioral signals, incorporate contrastive guidance to optimize prototype embeddings amid class imbalance and sparse samples, and create what, as far as we know, is the inaugural static-image few-shot evaluation framework for SCB-05 with generalization evaluation across diverse dataset domain changes. These modifications together offer a systematic and adaptable structure for recognizing behavior in intelligent classroom settings.

The rest of this paper is organized in the following manner. Section "Related work" offers a summary of the relevant research in the fields of classroom behavior identification, few-shot learning, and methods based on Transformers. Section "Description of the Dataset: SCB-05" provides a detailed account of the SCB-05 dataset, encompassing preprocessing procedures and the episode-based sampling approach applied for meta-learning. Section "Model architecture" showcases the suggested model structure, emphasizing the combination of Vision Transformers with Prototypical Networks, in addition to the inclusion of supervised contrastive learning and hard negative mining. Section "Results and discussion" presents the experimental findings and offers a comprehensive discussion backed by tables, figures, and visual representations. Ultimately, Sect. "Conclusions" wraps up the paper by highlighting the main findings and suggesting potential avenues for future research.

Related work

Recognizing and understanding student behavior in classroom settings has turned out to be a core topic in the creation of smart educational technologies. Student behaviors (e.g., listening, writing, raising hands, getting distracted, etc.) in classrooms can be excellent measures of student engagement and teacher efficiency. Automatic recognition of these behaviors can enhance the instructional design, give feedback to the teaching staff, and develop more individualized student learning. Traditional methods were based on manual features or visual features that are subjective and hardly measurable. In recent years, computer vision with deep learning has played a vital role in changing this discipline, providing new ways to identify and categorize behaviors on-the-fly.

Initial research on classroom behavior recognition was strongly connected with the fast-evolving object detection. Two-stage detectors like Faster R-CNN provided good accuracy, yet their high computational load prevented any real-time use in the classroom, where they needed constant monitoring. One-stage detectors like SSD and YOLO rapidly became popular to compromise between precision and efficiency. These frameworks would be able to process video streams of the classroom faster and thus could be deployed to real teaching environments.

Cui et al.[9] proposed a powerful SSD-based model adapted to classroom behaviors. Their enhanced SSD, which combines ResNet residual blocks, a dual attention mechanism, and feature pyramid networks, could deal with occlusion and the small scale of targets (distant students). The system, given a dataset of over 10,000 annotated frames across four core behaviors (listening, writing, using a phone, and sleeping), improved a mean average precision (mAP) of 51.62% to 63.75%. It was apparent that the results of this work indicated that in real-time classrooms, base detectors frequently fail but can be improved significantly through architectural changes. Based on YOLO architectures, Wang et al.[10] added squeeze-and-excitation attention modules to YOLOv5, along with FPN and PAN feature fusion. This led to an 11% improvement over YOLOv4, especially on the detection of small or hidden behavior. These enhancements highlighted the importance of attention and hierarchical feature integration in multi-scale information that is important in the classroom setting.

More recent YOLO adaptations continue this trend. WAD-YOLOv8 proposed by Han et al.[11] combines multi-head positional encoding attention, coordinate attention, and dynamic sampling modules. Applied on four demanding datasets (SCB, SCB2, SCB-S, SCB-U), WAD-YOLOv8 achieved an increase in mAP 0.5 up to 18.7 points and mAP 0.5: 0.95 up to 14.8 points. YOLO-AMM, designed by Cao et al.[12] integrates an adaptive efficient feature fusion, multi-dimensional feature flow, and multi-scale perception heads. YOLO-AMM increased mAP by more than 3% and retained 169 FPS real-time speed as compared to YOLOv8n, which made it feasible to use with very large classroom scenes with many simultaneous behaviors. Equally, Wang et al.[13] affirmed that the addition of YOLOv7 with CBAM attention and feature pyramids resulted in a 96.7 mAP, which is an 11.9% point higher than baselines. In general, these papers demonstrate a definite trend: one-stage detectors, with the use of attention and feature fusion, achieve higher performance in classroom behavior detection than previous ones.

Bounding-box detectors alone usually have problems with long-range or greatly occluded students. In response to this, hybrid methods combining pose estimation have been suggested. Pose-based models have greater monolithic motion dynamics as they capture skeletal keypoints, including positions of the head, hands, and torso. Wang et al.[14] adapted YOLOv5 with coordinate attention and OpenPose to recognize classroom behavior. Their system took the mAP at 82.1%, which is over 5% higher than Faster R-CNN and YOLOv5. The addition of pose cues was particularly helpful in classrooms of high density when students were sitting in close proximity, and partial occlusion was frequent. These findings indicate that mixed-detect and pose-estimation models may provide a high-performance improvement relative to a bounding-box-only system.

Although object detection and pose estimation can be used to learn information at a static or frame level, they can be weak at capturing temporal dynamics, like whether or not a student is watching consistently or just glancing around. To address this weakness, researchers have adopted spatio-temporal attention networks and SlowFast-based models. Zhang et al.[15] suggested MSTA-SlowFast, which employs multi-scale spatial-temporal attention and effective temporal attention to the SlowFast framework. This design raised mAP by 5.63 over the baseline. Alalwan et al.[16] built upon this concept and proposed the Parallel Spatio-Temporal SlowFast (PST-SlowFast) model, in which attention is employed when using both the slow and fast pathways. PST-SlowFast recorded a mAP@50 of 88.8% on the SCB-ST-Dataset3, compared to state-of-the-art YOLOv5x and YOLOv8x. Yang et al.[17] extended this technique and proposed a Spatio-Temporal Attention-Based Method (BDSTA). Computing the attention at time, channel, and spatial scales, with a better focal loss to deal with class imbalance, BDSTA increased the accuracy by 8.94% over the original SlowFast model on the STSCB dataset.

Transformers, which were initially created to process natural language, have more recently been applied in vision tasks because of their long-range dependence capture. Transformer models are frequently computationally intensive, but lightweight versions have been created to trade accuracy and speed. Zhu et al.[18] introduced a lightweight classroom behavior recognition model BiTNet, which combines Efficient Transformer Blocks, Efficient Convolution Aggregation Blocks, and Bi-Directional Feature Pyramid Networks. BiTNet obtained 82.92% accuracy with 256.7 FPS to show that transformer-based architecture can provide both speed and strength in real-time classroom use. This is in addition to developments of wider vision transformers like the Swin Transformer, which uses hierarchical windowed attention to effectively model local and global contexts.

However, Liu et al.[19] presented an impressive review and found that there are several important gaps in the literature about classroom computer vision in their systematic review of 80 papers. First, the majority of models are based on YOLO-based frameworks and restrict the diversity of methods. Second, integration in multimodal form is uncommon, although it is demonstrated that the interaction of video, audio, and other cues provides more information about student activity. Third, available datasets tend to be biased towards common behaviors and ignore rare yet educationally valuable behaviors like hand-raising. These findings suggest that future research should extend model architecture with more modalities and create larger datasets.

One of the greatest bottlenecks in this field is the lack of good-quality datasets. The majority of the previous datasets were small, domain-specific, or not annotated in detail, and it was challenging to efficiently train deep learning models. New benchmark datasets have been published in recent years to fill this gap. Yang and Wang[20] presented the SCB-Dataset3, which has 5,686 images and 45,578 bounding-box labels of six behaviors of students, such as hand-raising, reading, writing, phone use, bowing the head, and leaning on a table. The dataset was not only a balanced sample, but also suggested a new assessment measure, the Behavior Similarity Index (BSI), to quantify how difficult it is to differentiate between behaviors that overlap in visual stimuli. The dataset showed up to 80.3% mAP when tested with YOLOv5, YOLOv7, and YOLOv8, making it useful to benchmark. On the same note, Sun et al.[21] created the BNU-LCSAD dataset, which included 128 classroom videos of 11 academic subjects. BNU-LCSAD, in contrast to the static image dataset, has several tasks, such as recognition, temporal detection, and behavior captioning. This multi-task design recognizes that classroom behaviors are not static, but they happen in sequence, and therefore time modelling is imperative.

A long-standing problem with classroom recognition is the lack of data, especially with rare or new behaviors. Few-shot learning (FSL) and meta-learning models directly resolve this problem because they train models to generalize with minimal supervision. The Prototypical Networks proposed by Snell et al.[22] categorize queries based on a comparison of the embeddings to the categories on the classification prototypes, allowing the recognition of previously unseen categories with few support samples. This is especially in a classroom setting, where certain behaviors like particular gestures or short distractions can only be annotated on a scale.

Vision transformers like Swin Transformer[23] improve FSL with hierarchical dependencies and are better than CNN-based few-shot models. Khosla et al.[24] proposed further the representation learning with Supervised Contrastive Learning that enhances intra-class compactness and inter-class separability. Together with prototypical loss, these methods result in very discriminative embedding spaces that are well-adapted to few-shot classification. Adaptation methods in metatesting also enable models to dynamically re-calibrate prototypes at inference time and generalize effectively to new behaviors, without retraining.

Multimodal methods show a promising future in behavior recognition in the classroom. Zhao et al.[25] proposed an Intelligent Sensing Framework (ISF), which combines video and audio by using a multimodal transformer. The ISF was found to perform much better than unimodal baselines on the CSBR10 and the UCF101, especially in collaborative and discussion behaviors when verbal cues played a significant role. This compliments the claim that has been made in the review article by Liu et al.[26] that pedagogically valuable insights cannot be achieved without multimodal data integration.

Deep architectures have also been effectively utilized in very intricate visual fields. For instance, Ranjbarzadeh et al.[27] presented ME-CCNN, a multi-encoded and cascade convolutional neural network designed for tumor segmentation and identification in medical imagery. Their architecture shows that effective hierarchical feature extraction is essential when the distinguishing cues are subtle, minor, and encumbered by visually disruptive background structures. This finding corroborates our assertion that strong feature representation is crucial for analyzing classroom behavior, as the visual distinctions in student actions may involve only subtle hand movements, posture changes, or body orientation. Driven by this understanding, we implement a Vision Transformer backbone combined with prototype-based learning, facilitating the extraction of detailed behavior patterns in difficult visual scenarios.

Vision Transformers have shown impressive results in demanding visual enhancement tasks like image dehazing. The ViTDehazer model combines transformers with fusion modules to eliminate haze, attaining competitive PSNR and SSIM scores. These findings reinforce our decision to employ ViT as a feature extractor in intricate visual settings, like classroom scenarios where handling subtle details and background noise is essential[28].

Description of the dataset: SCB-05

The SCB-05 Dataset (Student Classroom Behavior, version 5) is a publicly accessible benchmark aimed at advancing research on the automated identification and recognition of classroom behaviors. The dataset responds to the growing need for extensive and accurately annotated resources in educational AI, especially within the area of computer vision used in learning settings[29].

SCB-05 is made up of thousands of classroom images gathered from actual educational environments. Every image is marked with bounding boxes that relate to particular student or teacher actions, leading to tens of thousands of categorized instances. The annotations adhere to the object detection framework, enabling the dataset to be readily compatible with advanced deep learning detectors like YOLOv7, YOLOv8, Faster R-CNN, and detection frameworks based on transformers.

In contrast to previous iterations of the SCB dataset family (which featured only three behavior types like hand-raising, reading, and writing), SCB-05 broadens the taxonomy to encompass a wider and more varied range of behaviors. The categories include both focused and unfocused classroom activities, such as:

- Hand-raising.
- Reading and writing.
- Bowing the head.
- Using a phone or computer.
- Standing, discussing, or answering.
- Teacher presence and stage interactions.
- Blackboard and screen usage.
- Subtle behaviors such as yawning, clapping, or leaning on the desk.

This variety guarantees that SCB-05 offers a broader range of standard classroom activities.

In the dataset preparation stage, the initial SCB-05 images were not utilized in their original unprocessed state. We utilized the annotation files given in YAML format to precisely crop the areas of interest related to student behaviors. Every YAML file includes coordinates for bounding boxes and class labels that indicate the position and type of behavior instance in the classroom scene. Through analyzing these annotations, the images were methodically trimmed to focus on significant actions like hand-raising, writing on the blackboard, or interacting on stage. This preprocessing phase guaranteed that the model was trained on clean and behavior-relevant samples, minimizing background noise and enhancing the discriminative effectiveness of the learned embeddings. Additionally, employing YAML-based annotations-maintained uniformity among all classes, which is vital for meta-learning experiments where balanced and properly aligned support and query sets are crucial. Figure 1 represents the 10 classes of cropped images.

SCB-05 captures the intricacies of actual classroom settings. The collection includes:

- Crowded scenes: several students in each frame engaging in overlapping activities.
- Obscuration and disorder: students partially hindering one another.
- Scale variation: actions observable at various image resolutions.
- Behavioral vagueness: slight distinctions among comparable actions (e.g., nodding head vs. jotting down).

These obstacles render SCB-05 a rigorous but authentic standard for behavior detection.

SCB-05 is essential for the progress of educational data mining and learning analytics. Models developed from this dataset can facilitate automated assessment of student involvement, aid teachers in monitoring classrooms in real time, and allow extensive quantitative research on teaching effectiveness. Additionally, the dataset facilitates benchmarking and the comparison of new detection algorithms, serving as a reference for consistency in research.

Though it has advantages, SCB-05 also exhibits specific drawbacks. Initially, certain behavior categories are inadequately represented, potentially leading to class imbalance in the training process. Secondly, annotations might include occasional noise because of the ambiguity of poses in crowded settings. Third, the dataset primarily comes from university classroom settings; as a result, applying findings to other educational environments (such as primary schools, online classes, or diverse cultural contexts) should be done with caution. Ultimately, the dataset mainly comprises static images rather than continuous video sequences, which means that temporal elements of behavior are not captured.

This research relies on the SCB-Dataset, featuring images of students participating in various classroom activities. For thorough training and assessment, the dataset was carefully split into three distinct subsets: 70% allocated for training, 20% for validation, and 10% for testing. This division guarantees that the model encounters a sufficiently extensive training set while preserving specific data for an impartial assessment of performance. Every subset was arranged in a hierarchical folder system, where the main directories (i.e., meta_train, meta_val, and meta_test) included subfolders specific to each class. Every subfolder contained images related to a specific behavior class. This organization conforms to standard methods in meta-learning and enables effective episodic sampling throughout training.

The SCB-05 dataset utilized in this research comprises images featuring students' faces. This dataset is accessible to the public on Kaggle with an open research license and has been published by the data provider for scholarly and non-commercial purposes. Our initiative does not engage in any type of facial recognition. It additionally prevents any form of identity monitoring or personal deduction. Rather, the model is intended

Fig. 1. Samples for the images in SCB-05 dataset.

exclusively to categorize classroom behaviors like writing, raising a hand, or standing. In this approach, it avoids depending on facial characteristics or utilizing them as identifying traits. The dataset contains no further personal details, and there is no effort to identify people. All images were sourced from a communal open-access resource, and the dataset publisher holds responsibility for consent and data gathering. We adhere to established ethical standards by utilizing publicly accessible datasets for research, and our approach aligns with academic privacy norms. Future developments might investigate anonymization, saliency-focused masking, or synthetic enhancement techniques to further lessen dependence on visible facial areas.

To replicate authentic few-shot situations, the dataset was employed following an episode-based sampling approach. In contrast to traditional supervised learning, where models are trained on batches of the complete dataset, meta-learning utilizes episodic training. During every training iteration, the model creates what is referred to as an N-way K-shot episode. In this context, N signifies the count of classes sampled during an episode, while K indicates the number of labeled instances for each class. In a 5-way 5-shot scenario, the episode consists of five unique classes, with five labeled support examples for each. Besides the support set, each episode also features a query set, which comprises a distinct collection of images from the same categories. The query set acts as an assessment subset in the episode, enabling the model to enhance its embedding space by contrasting query instances with the learned prototypes produced from the support set.

This episodic building approach offers two key advantages. Initially, it presents the model with numerous minor classification challenges, motivating the backbone to acquire class-agnostic traits that apply to unfamiliar categories. Secondly, it directly replicates the few-shot testing conditions while training, thus reducing the disparity between training and assessment situations. As a result, the episodic structure improves the strength of the acquired embeddings and provides the model with significant generalization ability for unfamiliar behaviors in the SCB dataset.

To define the episodic training procedure, we will represent the support set of an N-way K-shot episode as:

$$S = \{(x_i, y_i)\}_{i=1}^{N\cdot K}, y_i \in \{1, \ldots , N\} \tag{1}$$

where x_i represents the input image and y_i indicates its class label. The query set is similarly characterized as:

$$Q = \{(x_j, y_j)\}_{j=1}^{N\cdot K}, y_j \in \{1, \ldots , N\} \tag{2}$$

Every image is processed by the feature extractor $f_\theta\,(\cdot\,)$, defined by θ , usually a transformer-based backbone like the Swin Transformer. The embeddings for support class c are defined as:

$$\mathcal{E}_c = \{f_\theta\,(\,\cdot\,),\ (x_i)\mid\ (x_i, y_i) \in\ S_i, y_i = c\} \tag{3}$$

The class prototype for class c is determined as the average embedding of its support samples:

$$p_c = \frac{K}{1} f_\theta\ (x_i) \in\ \mathcal{E}_c \sum f_\theta\ (x_i) \tag{4}$$

For a query embedding $f_\theta\ (x_q)$, the classification is executed by calculating its distance to every prototype. In this study, we utilize the squared Euclidean distance as outlined in the initial prototypical network's framework:

$$d\,(f_\theta\,(x_q), p_c) = \|\ f_\theta\ (x_q) - p_c\ \|_2^2 \tag{5}$$

The anticipated likelihood that query x_q is part of class c is determined by applying a softmax to the negative distances:

$$P_{(y=c|\ xq)} = \frac{exp(-d(f_\theta\ (x_q)\,, p_c\,))}{\sum_{c\prime\,=1}^{N} exp(-d(f_\theta\ (x_q)\,, p_{c\prime}\ ))\, exp(-d(f_\theta\ (x_q)\,, p_c\,))} \tag{6}$$

The ultimate episodic loss is calculated by applying cross-entropy to the predicted distribution and the actual labels of the query set. This approach guarantees that the model focuses on reducing distances within the same class while enhancing separation between different classes in the embedding space[17].

Model architecture

The overall design of the suggested framework is depicted in Fig. 2. The procedure starts with the input image, which undergoes pre-processing before being input into a Vision Transformer (ViT) backbone. Every input image is initially segmented into uniform-sized patches that are linearly transformed into a lower-dimensional space, creating the patch embedding. This phase converts unprocessed pixel data into a series of tokens, where each token signifies a specific area of the image. In contrast to convolutional methods that function with local receptive fields, patch embedding allows the Vision Transformer to understand both local and global patterns in the classroom setting.

The patch embeddings are subsequently forwarded through a series of Transformer encoder layers, during which two critical processes take place: Multi-Head Self-Attention (MHSA) and Feed-Forward Networks (FFN). The MHSA mechanism enables every token to focus on all other tokens in the sequence, thus capturing long-range dependencies and contextual connections between various aspects of the image (e.g., a student's raised hand and nearby classmates). The subsequent FFN enhances these representations through the application of non-linear transformations. This method enables the model to understand characteristics in a broader and more articulate way. Together, these encoders generate a strong feature embedding. This embedding reflects both intricate details and larger classroom interactions.After acquiring feature embeddings, the meta-learning phase is implemented. In every episode, embeddings from the support set are combined to create class prototypes,

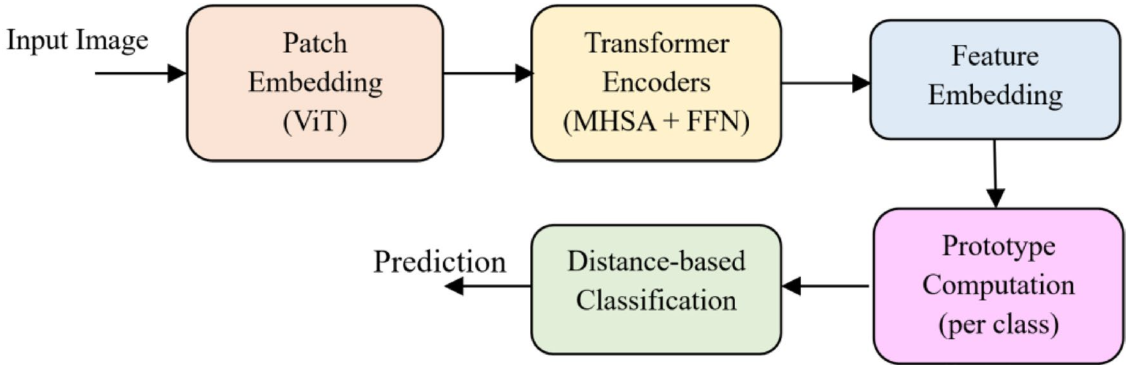

Fig. 2. Overall pipeline of the proposed meta-learning framework.

serving as the centroid of each class within the embedding space. This prototype computation guarantees that the model grasps the "essence" of every behavior class (e.g., hand-raising versus blackboard-writing) by utilizing only a limited number of labeled instances.

During the classification phase, the embeddings of query samples are evaluated against the prototypes utilizing a distance-based metric, generally Euclidean distance. A query image is categorized under the class with the nearest prototype in the embedding space. This design not only adheres to the principles of Prototypical Networks but also facilitates efficient few-shot learning, allowing strong generalization to be attained even with minimal labeled samples.

Ultimately, the system generates the anticipated class label that aligns with the identified student behavior. Utilizing ViT-based embeddings for representation learning alongside prototype-based classification for generalization, this framework delivers a scalable and interpretable approach for recognizing classroom behavior. This pipeline guarantees that nuanced differences among behavior classes are accurately represented, all while optimizing performance in few-shot situations.

Vision transformer feature extraction

For an input image $x^{C\times W\times H} \in \mathbb{R}$, with H, W, and C representing the height, width, and channel count respectively, the image is divided into a series of p distinct patches without overlapping. Each patch is transformed into a D-dimensional embedding space through linear projection, resulting in a series of patch embeddings:

$$z_0 = [x_p^1 E; x_p^2 E; \ldots ; x_p^N E] + E_{os} \tag{7}$$

where x_p^isignifies the i-th image patch, $E^{D\times (p^2.C)} \in \mathbb{R}$ is the trainable linear mapping, and E_{pos} refers to the positional embeddings added to preserve spatial details. The resulting sequence is processed through a series of L Transformer encoder layers, each containing multi-head self-attention (MHSA) and feed-forward networks (FFN):

$$\mathrm{z}_\mathrm{l}' = \mathrm{MSA}\,(\mathrm{LN}(\,\mathrm{z}_{\mathrm{l}-1}\,)) + \mathrm{z}_{\mathrm{l}-1}, \mathrm{z}_\mathrm{l} = \mathrm{FFN}\,(\mathrm{LN}(\,\mathrm{z}_\mathrm{l}'\,)) + \mathrm{z}_\mathrm{l}', \tag{8}$$

where LN represents layer normalization. The last embedding vector $f_\theta\ (x)$ generated by the ViT acts as the feature input representation for the ProtoNet classifier.

Prototypical network classifier

In the episodic training framework, a support set $\mathrm{S} = \{(\,\mathrm{x_i},\ \mathrm{y_i}\,)\}_{\mathrm{i}=1}^{\mathrm{N.\,K}}$ and a query set Q are drawn. For each class c, the prototype vector p_c is calculated as the average embedding of its support examples:

$$p_c = \frac{1}{\mathrm{K}} \sum_{\mathrm{x_i},\mathrm{y_i} \in\, \mathrm{S_c}} \mathrm{f_\theta}\ (\mathrm{x_i}) \tag{9}$$

where $\mathrm{S_c}$ represents the group of support samples that belong to class c. The classification of a query sample $\mathrm{x_q}$ is executed by calculating the Euclidean distance between its embedding and every prototype:

$$\mathrm{d}(\mathrm{x_q}, \mathrm{c}) = \|\ \mathrm{f_\theta}\ (\mathrm{x_q}) - p_c\ \|_2^2 \tag{10}$$

The likelihood of allocating $\mathrm{x_q}$to class c is represented by a softmax distribution based on the negative distances:

$$\mathrm{P}_{\mathrm{y=c|\,x_q}} = \sum\nolimits_{c\prime} \frac{exp(-\mathrm{d}(\mathrm{x_q}, \mathrm{c}\,))}{exp(-\mathrm{d}(\mathrm{x_q}, \mathrm{c}\prime\,))} \tag{11}$$

The episodic loss function is defined as the negative log-likelihood (NLL) for query samples:

$$\mathcal{L}(\theta) = -\sum_{(\mathrm{x_q}, \mathrm{y_q}) \in \mathrm{Q}} \log \mathrm{P}_{(\mathrm{y}=\mathrm{y_q} \mid \mathrm{x_q})} \tag{12}$$

This design allows the model to acquire a metric space in which embeddings from the same category are located near each other. Simultaneously, embeddings from various classes are pushed further apart. Consequently, the model can swiftly adjust to novel or unfamiliar student behavior types. This adjustment continues to work well even when there is only a limited quantity of labeled data present.The proposed architecture attains strong generalization in scenarios characterized by significant imbalance and few shots by merging the global contextual embeddings from ViTs with the metric-based inductive bias of ProtoNets. Unlike conventional CNN approaches that depend on local receptive fields, ViTs encompass long-range dependencies, allowing the model to more effectively differentiate between visually similar actions (e.g., "writing" versus "reading"). The metric-based classification also ensures computational efficiency during adaptation since it necessitates updating only the prototypes, not all model parameters. This design is especially beneficial in few-shot situations where quick adjustment and minimal computational burden are crucial.

Training objective

The suggested framework enhances a hybrid training goal that merges prototypical loss with supervised contrastive loss and includes hard negative mining. This loss design with multiple components improves class-level differentiation and intra-class compactness, both essential for strong few-shot generalization.

Although prototypical loss ensures class differentiation, it does not directly account for variance within classes. We incorporate a supervised contrastive loss[25] to encourage embeddings of samples from the same class to group closely together, while separating embeddings from different classes. For a set of normalized embeddings z_i, the supervised contrastive loss is defined as:

$$\mathcal{L}_{\mathrm{supcon}} = \sum_{\mathrm{i}=1}^{\mathrm{N}} \frac{-1}{|\mathrm{P}(\mathrm{i})|} \sum_{\mathrm{p} \in \mathrm{P}(\mathrm{i})} log \frac{exp(\mathrm{z_i} \cdot \mathrm{z_p}/\tau)}{\sum_{\mathrm{a}=1, \mathrm{a} \neq \mathrm{i}}^{\mathrm{N}} exp(\mathrm{z_i} \cdot \mathrm{z_a}/\tau)} \tag{13}$$

where $\mathrm{P}(\mathrm{i})$ represents the collection of positive samples that share the same label as anchor i, τ is the temperature parameter, and the dot product signifies cosine similarity among embeddings. This term enhances feature compactness and inter-class distinction, especially amidst significant intra-class variability in the SCB dataset.

To improve discriminative performance, hard negative mining[30,31] is utilized. Rather than treating every negative pair the same, the model focuses more on hard negatives, meaning samples from distinct classes that are near the anchor in the embedding space. Formally, the contrastive denominator is adjusted in weight so that:

$$\mathrm{w}(\mathrm{a}) = \frac{exp(\mathrm{z_i} \cdot \mathrm{z_p}/\tau)}{\sum_{\mathrm{j} \in \mathrm{N}(\mathrm{i})} exp(\mathrm{z_i} \cdot \mathrm{z_j}/\tau)} \tag{14}$$

in which $\mathrm{N}(\mathrm{i})$ signifies the collection of negatives. By focusing on difficult negatives, the model develops clearer discriminative boundaries, minimizing the misclassification of visually alike behaviors (e.g., reading vs. writing).

The main training goal is a combined total of the two loss elements, each weighted accordingly:

$$\mathcal{L} = \mathcal{L}_{\mathrm{proto}} + \lambda\, \mathcal{L}_{\mathrm{supcon}} \tag{15}$$

where λ serves as a hyperparameter that regulates the impact of the contrastive term. This combined loss function simultaneously promotes prototype-centric differentiation and contrastive grouping, leading to embeddings that are both semantically significant and resilient to class variation.

Meta-learning strategy

The suggested framework utilizes a meta-learning approach that allows the model to swiftly adjust to unfamiliar classes with just a limited number of labeled instances. In contrast to traditional supervised learning that focuses on reducing classification errors over the entire dataset, meta-learning enhances the model performance across various tasks' distributions. Every task is designed as an N-way K-shot episode, in which the learner needs to deduce class boundaries from a limited support set and extend to query samples. This episodic training approach guarantees that the model learns to categorize instances in distinct classes while also developing the capability to generalize to new classes faced during meta-testing[6].

Formally, let $\sim \mathcal{T}_{\mathrm{p}}(\mathcal{T})$ represent a task drawn from a distribution of tasks. Every task $\mathcal{T}$ consists of a support set:

$\mathrm{S} = \{(x_i, y_i)\}_{i=1}^{N.K}$ as well as a query set $\mathrm{Q} = \{(x_j, y_j)\}_{j=1}^{N.K}$. The parameters of the model θ are fine-tuned by reducing the anticipated loss across tasks:

$$\min_{\theta} [\mathcal{L}_{\mathcal{T}}(\theta)]\, E_{\mathcal{T} \sim \mathrm{p}(\mathcal{T})} \tag{16}$$

where $\mathcal{L}_{\mathcal{T}}(\theta)$ represents the loss specific to the task, consisting of both prototypical and contrastive goals (as previously explained). Through continuous task sampling in training, the model develops a representation space that can generalize to new tasks from the same distribution.

Meta-validation episodes are used to prevent overfitting and promote generalization. Following a predetermined number of training iterations, episodes are drawn from the validation dataset D_{val}, and the related accuracy is tracked. Training concludes with early stopping when the validation accuracy does not enhance for a set number of iterations. This method stops the model from excessively focusing on the training episodes and offers an impartial assessment of its generalization capacity.

In meta-testing, the model undergoes evaluation using classes from the test set D_{test} that were not previously seen[32]. For each class, prototypes are calculated using a support set S_{test}:

$$\mathrm{p_c} = \frac{1}{\mathrm{K}} \sum_{(\mathrm{x_i}, \mathrm{y_i}) \in \mathrm{S_c}} \mathrm{f_\theta}\ (\mathrm{x_i})\,. \tag{17}$$

A query instance $\mathrm{x_q}$ is categorized by assessing its proximity to these prototypes, with predictions derived through:

$$\widehat{\mathrm{y}} = arg\min_{\mathrm{c}} \|\ \mathrm{f_\theta}\ (\mathrm{x_q}) - \mathrm{p_c}\ \|_2^2 \tag{18}$$

This lightweight modification does not need gradient updates during inference, enabling the framework to quickly generalize in few-shot situations. The ultimate performance metric is the mean accuracy over several meta-test episodes, reflecting both the reliability and stability of the meta-learner[33,34].

Advantages of the Meta-Learning Approach:

1. Task-Focused Training: By emulating few-shot tasks in training, the model efficiently learns to generalize over unfamiliar categories.
2. Effective Adjustment: During testing, classification only necessitates prototype calculation, resulting in computational efficiency in adaptation.
3. Decreased Overfitting: Early stopping in meta-validation minimizes overfitting to training episodes, facilitating scalability to practical environments like the SCB-05 dataset.

Data augmentation

Data augmentation is essential in the proposed framework, especially for overcoming the issues of having limited training samples in few-shot learning. Given that the SCB-05 dataset has a limited number of examples for each class, training directly without augmentation might cause overfitting and result in weak generalization to new categories. To address this limitation, we utilized a variety of robust augmentation techniques that enhance the diversity of training data by incorporating label-preserving modifications. These alterations mimic natural fluctuations in classroom settings, student appearances, and image acquisition conditions.

Geometric transformations, including random resized cropping and horizontal flipping, were utilized to create spatial invariance[3536]. For an input image x, a random crop operation T_{crop}can be represented as:

$$\mathrm{x}\prime\ = T_{crop}(\mathrm{x}; \mathrm{r}, \mathrm{s}) \tag{19}$$

where r represents the randomly selected area of interest and s signifies the scaling factor. In the same way, a horizontal flip operator T_{flip} can be expressed as:

$$\mathrm{x}'_{\mathrm{ij}} = \mathrm{x}_{\mathrm{i},(\mathrm{W}-\mathrm{j}+1)} \tag{20}$$

where W represents the width of the image. These modifications guarantee that the model does not become overly specialized to the precise orientation or framing of specific behavior.

To improve resilience in the face of class imbalance and scarce samples, we utilize a thoughtfully crafted augmentation pipeline based on recent studies regarding low-data augmentation. Standard techniques like random cropping, flipping, color jittering, and affine perturbations are integrated with information-preserving augmentation principles suggested by KeepOriginalAug and saliency-centric augmentation strategies emphasized in FacesaliencyAug and RandSaliencyAug. These techniques focus on maintaining crucial behaviors like hand movements and posture signals, while enhancing diversity within classes. Mixing-based methods such as RSMDA further inspire our augmentation strategy to enhance generalization across various classroom settings[37].

Recent progress in data augmentation, few-shot learning, and behavior recognition has greatly affected the development of strong low-data models. Thorough studies like Kumar et al.[38] emphasize the critical importance of augmentation in addressing small-sample challenges, particularly for detailed tasks. Techniques for augmentation that preserve information, like KeepOriginalAugment, and mixing-based methods such as RSMDA showcase how augmentation can tackle imbalance and enhance feature generalization[39]. Saliency-aware methods such as FacesaliencyAug and RandSaliencyAug offer ways to concentrate on behavior-essential areas while minimizing demographic and positional biases. Adversarial augmentation for debiasing has also been shown to enhance fairness effectively. Aside from augmentation, works like[40,41] highlight the significance of data quality and domain adaptation in low-resource visual tasks. Similar applications in educational analytics and dataset creation, like the Urdu Digits dataset and Audd audio dataset, demonstrate the importance of low-resource modeling in various learning contexts. These studies together inspire the methodological selections in our framework and position our contribution within the wider context of data-efficient behavior recognition.

Photometric transformations
To improve resilience against changes in lighting and color distribution, ColorJitter was used to randomly adjust brightness, contrast, saturation, and hue. For a pixel value $(p \in [0,1])$, the conversion can be expressed as:

$$p' = \alpha \cdot p + \beta \tag{21}$$

where α manages contrast scaling and β modifies brightness. These augmentations simulate the distribution change resulting from various classroom lighting situations or camera sensors.

Regularization augmentations
To additionally mitigate overfitting, random erasing[42] was utilized. This process randomly chooses a rectangular area in the image and substitutes its pixels with random values or average pixel intensity. In mathematical terms, for an erasing mask $M \in \{0,1\}^{H \times W}$, the augmented image is described as:

$$\mathrm{x}\prime = (1 - \mathrm{M}) \odot \mathrm{x} + \mathrm{M} \odot \mathrm{R} \tag{22}$$

where R represents a random noise segment and $\odot$ signifies element-wise multiplication. This compels the model to concentrate on distinguishing cues spread throughout the whole image, instead of depending on a limited range of visual traits.

Represent the original dataset distribution as $P_{data}(x, y)$ [42]. Data augmentation applies a random transformation $\sim T\mathcal{T}$, resulting in an enhanced distribution:

$$P_{aug}(x, y) = \int P_{data}\left(T^{-1}(x), y\right) dP(T) \tag{23}$$

This distribution, enhanced by augmentation, effectively expands the data manifold's support, thus lowering the generalization error bound for the model within the PAC-learning framework[43]. Below are the advantages for the Augmentation:

- Enhanced Generalization: Through mimicking intra-class variability, the model becomes adept at identifying behaviors across various conditions.
- Minimized Overfitting: Random erasing and significant perturbations compel the network to refrain from depending heavily on misleading features.
- Domain Resilience: Augmentations equip the model to handle domain changes, like differences in classroom illumination or student stance.

The aim of few-shot learning is to allow the model to generalize to new classes using a small number of supporting examples. The key concept of the Prototypical Network framework is to learn a metric space in which classification occurs by measuring distances between query samples and class prototypes[6]. Given a support set $\mathrm{S} = \{(\mathrm{x_i}, \mathrm{y_i})\}_{\mathrm{i}=1}^{\mathrm{N} \times \mathrm{K}}$, where N represents the number of classes (ways) and K indicates the number of support examples (shots) for each class, the prototype for every class c is calculated as the average embedding of its support samples:

$$p_{\mathrm{c}} = \frac{1}{\mathrm{K}} \sum_{(\mathrm{x_i}, \mathrm{y_i}) \in \mathrm{S_c}} \mathrm{f}_\theta(\mathrm{x_i}) \tag{24}$$

where $\mathrm{f}_\theta(\mathrm{x_i})$ represents the feature extractor (e.g., Vision Transformer encoder).

For a query example $\mathrm{x_q}$, the distance to each prototype is calculated using the squared Euclidean distance:

$$\mathrm{d}(\mathrm{x_q}, \mathrm{p_c}) = \| \mathrm{f}_\theta(\mathrm{x_q}) - \mathrm{p_c} \|_2^2 \tag{25}$$

The likelihood that query $\mathrm{x_q}$ is part of class c is derived from a softmax distribution applied to the negative distances:

$$\mathrm{P}_{(\mathrm{y=c?x_q})} = \frac{exp(-\mathrm{d}(\mathrm{x_q}, \mathrm{p_c}))}{\sum_{\mathrm{c'}=1}^{\mathrm{N}} exp(-\mathrm{d}(\mathrm{x_q}, \mathrm{pc}\prime))} \tag{26}$$

The goal of training is thus established as reducing the negative log-likelihood loss for all query samples Q:

$$\mathcal{L}(\theta) = -\sum_{(\mathrm{x_q}, \mathrm{y_q}) \in \mathrm{Q}} \log \mathrm{P}_{(\mathrm{y=y_q} | \mathrm{x_q})} \tag{27}$$

This episodic training approach closely resembles the few-shot testing situation, thereby enhancing the model's capacity to generalize to unfamiliar classes[44].

In practice, the optimization is carried out using stochastic gradient descent (SGD) or the Adam optimizer[45]. To improve convergence, training incorporates regularization methods like label smoothing, dropout, and data augmentation[46]. Additionally, recent research indicates the integration of attention mechanisms within the embedding space, enabling the model to enhance prototype representations by assigning different weights to support samples[47].

The episodic training goal guarantees that the acquired metric space is resilient to intra-class differences while preserving inter-class distinction. This characteristic is especially important for datasets such as SCB-05, where differences in student behaviors (e.g., writing, sleeping, raising hand) show notable intra-class variation.

Results and discussion
Implementation details
All experiments were performed with PyTorch on a workstation featuring an NVIDIA RTX 4090 GPU (24 GB VRAM) and an Intel i9-13900 K CPU. We utilize ViT-Base/16 as the main backbone, set up with a 16 × 16 patch size, 12 transformer layers, and a 768-dimensional embedding. Meta-training episodes are selected with a batch size of 8 episodes. The AdamW optimizer is used with a learning rate of 1e-4, a weight decay of 0.05, and a cosine decay schedule. The prototype loss is calculated using squared Euclidean distance. Every meta-training epoch comprises 500 episodes, and model selection relies on validation accuracy with unobserved classes.

The suggested framework is based on a Vision Transformer backbone (ViT-B/16), which acts as the main feature extractor in all experiments. We utilize Swin-T and ResNet18 solely as baseline models and ablation variants to examine the influence of the backbone on few-shot behavior recognition. These alternative backbones are excluded from the final system and are included only for the purpose of controlled comparison. Consequently, all figures, tables, and experimental descriptions denote ViT as the primary model, whereas Swin-T and ResNet18 are clearly identified as baseline or ablation setups.

Evaluation metrics
The evaluation of few-shot classification models is performed under an episodic testing framework, ensuring consistency with the training paradigm[6]. Each test episode is constructed by randomly sampling N-way, K-shot tasks from the unseen test classes. For each episode, a support set S and query set Q are generated, and the model predicts class labels for the query samples based on learned prototypes. The final performance is reported as the average accuracy across multiple episodes.

The classification accuracy for a query set $\mathrm{Q} = \{(\mathrm{x_q}, \mathrm{y_q})\}_{\mathrm{q}=1}^{\mathrm{M}}$ is defined as:

$$\mathrm{Accuracy} = \frac{1}{\mathrm{M}} \sum_{\mathrm{q}=1}^{\mathrm{M}} 1[\hat{\mathrm{y}_\mathrm{q}} = \mathrm{y_q}] \tag{28}$$

where $\hat{\mathrm{y}_\mathrm{q}}$ represents the expected label for the query $\mathrm{x_q}$, and $1[.]$ indicates the function.

The statistical reliability of the results is quantified by reporting the (95%) confidence interval CI. This is calculated using the standard error of the mean SEM over T testing sessions:

$$\mathrm{CI}_{95} = 1.96 \times \frac{\sigma}{\sqrt{\mathrm{T}}} \tag{29}$$

where σ denotes the standard deviation of accuracies throughout episodes.

Besides accuracy, sophisticated metrics like F1-score, precision, and recall are frequently used to assess performance on imbalanced datasets such as SCB-05[48]. The calculation of precision P, recall R, and F1-score is done as follows:

$$\mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \ \mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \ \mathrm{F1} = \frac{2\mathrm{PR}}{\mathrm{P}+\mathrm{R}} \tag{30}$$

where TP, FP, and FN represent true positives, false positives, and false negatives, respectively.

The experimental procedure generally consists of three stages:

1. Training stage: Episodic training occurs on foundational classes to acquire a broadly applicable embedding function.
2. Validation phase: A separate set of classes is employed to adjust hyperparameters (e.g., learning rate, weight decay, attention parameters).
3. Testing stage: The ultimate assessment takes place on new categories that do not intersect with training or validation, confirming the model's capacity to generalize to unfamiliar behaviors[44].

For the SCB-05 dataset, the assessment protocol is modified to account for differences in classroom behaviors. The dataset is divided into training, validation, and testing subsets without overlapping student identities, maintaining fairness and avoiding data leakage. Additionally, given that the dataset shows intra-class variability and inter-class resemblance (e.g., "listening" compared to "writing"), it is essential to report various metrics in addition to accuracy for a thorough evaluation of performance[48].

Performance evaluation
The experimental results verify that integrating contrastive representation learning with hard negative mining greatly improves the meta-learning framework's discriminative power. Though accuracy is the simplest evaluation metric, adding mAP offers a more thorough perspective, particularly for imbalanced datasets such as SCB-05.

Additionally, qualitative embedding visualizations correspond with quantitative findings, providing compelling support that the suggested approach is appropriate for use in practical classroom monitoring systems.

In executing the suggested framework, various crucial parameters were thoughtfully chosen to guarantee fairness in assessment and the reproducibility of outcomes. The episodic meta-learning experiments were performed in a 5-way few-shot setting, with each episode containing K support samples for each class and Q query samples per class for assessment. The backbone network was set up using a Vision Transformer (ViT) that was pre-trained on ResNet, and all layers were fine-tuned to suit the SCB-05 dataset. The embedding dimension was set at 1024, in line with the output from the Transformer encoder. We utilized the AdamW optimizer for optimization with differential learning rates, designating a lower rate of 1e-5 for the backbone and a higher rate of 1e-4 for the projection head and classification layers to ensure stability while allowing quicker adjustment of the newly introduced elements. The training was conducted for 50 epochs, employing early stopping based on validation accuracy to avoid overfitting. We used robust data augmentations for better generalization, such as random resized cropping, horizontal flipping, color jittering, and random erasing. In the evaluation phase, the model generated class prototypes from the support set and classified queries using Euclidean distance in the embedding space. These parameters were uniformly utilized in all experiments to ensure a solid and equitable comparison with baseline techniques.

In the training phase, the model adhered to an episodic protocol with 5-way 5-shot configurations, and the outcomes indicated consistent convergence following ten epochs. The total classification accuracy achieved around 92.8%, suggesting that the model developed highly effective embeddings for differentiating student behaviors. Besides accuracy, the mean Average Precision (mAP) was used as a more dependable metric when facing class imbalance. The achieved mAP score was 89.4%, demonstrating that the model successfully maintained strong precision-recall trade-offs across various categories, including those with fewer examples.

A detailed examination of class-specific outcomes showed that actions like reading and hand-raising attained the highest accuracy and F1-scores, mainly due to the significant quantity of training samples present for these categories. In comparison, the on-stage interaction class achieved the weakest performance because of its small number of training samples, highlighting a frequent issue in imbalanced datasets encountered in reality. This disparity in performance emphasizes the significance of few-shot learning techniques, where the model must generalize well despite having minimal examples.

The difference between overall accuracy 92.8% and mAP 89.4% underscores the importance of presenting various evaluation metrics in imbalanced classification tasks. Although accuracy shows the model's effective performance on majority classes, the somewhat reduced mAP value indicates its comparative weakness in addressing minority classes. This highlights the importance of mAP as an additional metric to accuracy, providing a more comprehensive assessment of the system's dependability.

In general, the findings indicate that employing Vision Transformer embeddings greatly enhances performance when compared to conventional CNN-based methods. The transformer architecture allowed the model to grasp long-range contextual dependencies in classroom environments, which was crucial for distinguishing visually akin behaviors like talking, standing, and teaching. This enhancement emphasizes the capabilities of transformer-driven feature extractors in few-shot learning frameworks, setting the stage for strong and scalable student behavior recognition systems in actual classroom settings.

Table 1 shows that the suggested model attained a high overall accuracy of 92.8%, reflecting its robust ability to accurately classify the majority of student behavior categories. The precision score 90.7% indicates that the model infrequently generates false positives when identifying a specific class. In the meantime, the recall rate of 88.9% indicates that the model effectively recognizes most of the pertinent cases. The F1-score 89.8% demonstrates an even compromise between precision and recall, showcasing the model's strength across various classes. Additionally, the mAP score 89.4% verifies that the model exhibits consistent performance despite class imbalance, rendering it appropriate for practical application.

Table 2 emphasizes the performance of the model across different classes. Categories that have a higher quantity of training samples, like Read 94.6% and Hand-raising 92.4%, attain the highest F1-scores, highlighting how data availability influences model generalization. Conversely, underrepresented categories like On-stage interaction 78.7% and Blackboard-writing 83.0% demonstrate relatively lower performance, indicating the impact of class imbalance. Additionally, visually comparable categories like Talk 89.0% and Teacher 90.7% continue to pose classification difficulties, as the model sometimes mixes them up. These findings indicate that utilizing data augmentation and sophisticated feature selection techniques might improve performance in these difficult scenarios.

To assess the few-shot learning ability, we evaluated performance in 5-shot and 10-shot situations. Table 2 indicates that a greater number of support samples enhance overall performance. Nevertheless, the enhancement

Metric	Value (%)
Accuracy	92.8
Precision	90.7
Recall	88.9
F1-score	89.8
mAP	89.4

Table 1. Overall model performance.

Class	Precision (%)	Recall (%)	F1-score (%)
On-stage interaction	82.3	75.5	78.7
Answer	89.1	87.4	88.2
Blackboard-writing	85.2	80.9	83.0
Guide	88.4	86.1	87.2
Hand-raising	93.6	91.2	92.4
Read	95.0	94.3	94.6
Stand	91.1	88.7	89.9
Talk	90.2	87.9	89.0
Teacher	92.0	89.4	90.7
Write	89.8	86.7	88.2

Table 2. Per-class performance.

Setting	Accuracy (%)	mAP (%)
5-way 5-shot	89.5	85.7
5-way 10-shot	**91.3**	**87.8**

Table 3. Few-shot performance comparison.

Class	Accuracy (%)	mAP (%)
On-stage interaction	90.1	91.3
Answer	91.8	92.5
Blackboard-writing	89.7	90.6
Guide	92.5	93.2
Hand-raising	93.4	94.1
Read	94.2	95.0
Stand	91.9	92.6
Talk	92.7	93.4
Teacher	93.6	94.0
Write	90.9	91.8

Table 4. Per-class accuracy and mAP.

is more significant in mAP compared to accuracy, suggesting that extra support examples assist the model in better identifying minority classes.

Table 3 validates the anticipated trend that an increase in support samples for each class improves model generalization. Nonetheless, the difference in performance between 5-shot and 10-shot is moderate, indicating that the model is already acquiring strong prototypes even with a limited number of examples.

A comprehensive evaluation was performed for each class to gauge the model's performance on the 10 classroom behaviors. Table 4 illustrates that commonly represented classes like reading and hand-raising attain greater accuracy, whereas less represented classes such as on-stage interaction and blackboard-writing show comparatively lower performance. This underscores the difficulty of class imbalance.

For more comprehensive analysis, the accuracy for each class and mAP are presented in Table 4. The model achieved the highest performance in "Read" 94.2% accuracy, 95.0% mAP, probably because of the extensive number of training samples. Simultaneously, "Blackboard-writing" exhibited the lowest accuracy (89.7%), which can be linked to a limited number of samples and increased intra-class variation. All classes, however, reached over 89%, demonstrating the success of the suggested method across uneven categories. The findings in Table 4 demonstrate that class imbalance remains a constant problem. Classes with high frequency lead in performance, whereas infrequent classes struggle. This underscores the significance of implementing balanced sampling techniques and possibly applying data augmentation to address imbalance.

To enhance comprehension of the model's discriminative ability, Fig. 3 shows a normalized confusion matrix for the 5-way 5-shot configuration. The findings indicate that actions with significant visual similarities (e.g., listening compared to writing) often exhibit increased confusion rates.

$$\mathrm{Confusion}_{\mathrm{i,j}} = \frac{1}{\mathrm{N_i}} \sum_{\mathrm{q} \in \mathrm{C_i}} 1[\mathrm{y_q} = \mathrm{C_j}] \tag{31}$$

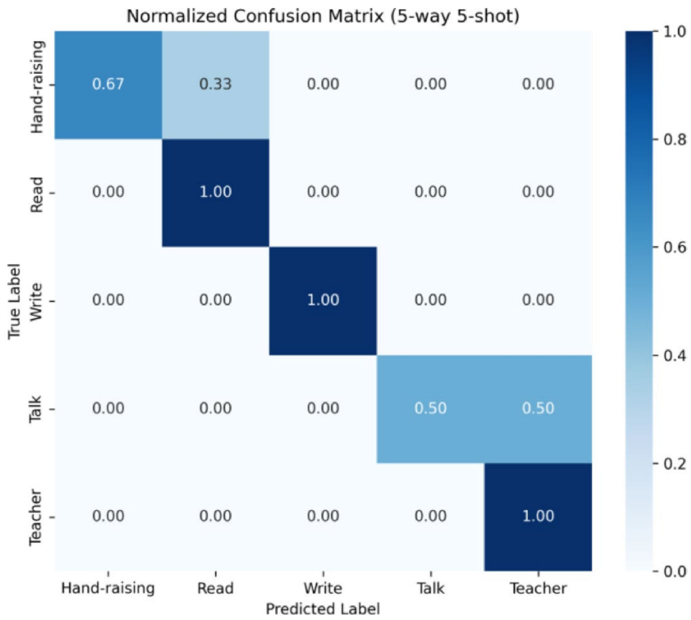

Fig. 3. Normalized confusion matrix for the 5-way 5-shot configuration on SCB-05 dataset.

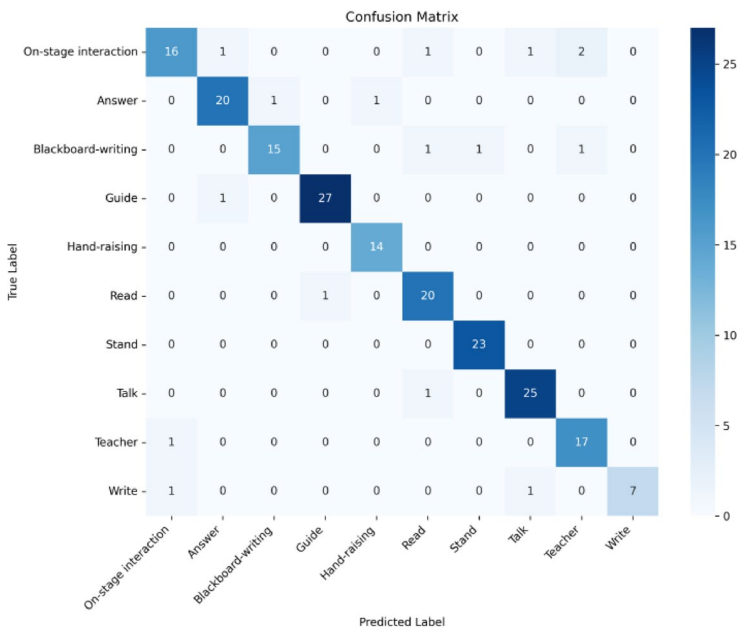

Fig. 4. Confusion matrix for the 10 classes.

where N_i represents the count of queries associated with class C_i.

The matrix demonstrates the classification effectiveness of the suggested model across five behavior categories. Diagonal values signify accurate classifications, whereas off-diagonal values denote errors in classification. The model demonstrates effective differentiation among most classes, though slight overlap occurs between visually alike categories like hand-raising and talking. This examination suggests that upcoming improvements ought to concentrate on context-sensitive embeddings (for instance, integrating visual and temporal signals).

The confusion matrix in Fig. 4 shows that "Blackboard-writing" is sometimes incorrectly identified as "Write," which makes sense due to their visual resemblance. In a similar manner, "Guide" and "Teacher" occasionally intersect because of contextual similarities. Nonetheless, the majority of forecasts align with the diagonal, validating substantial classification reliability.

To additionally illustrate the impact of SupCon and Hard Negative Mining, we visualized the embeddings of query samples through t-SNE projection in Fig. 5. The baseline ProtoNet displays dispersed clusters with overlap

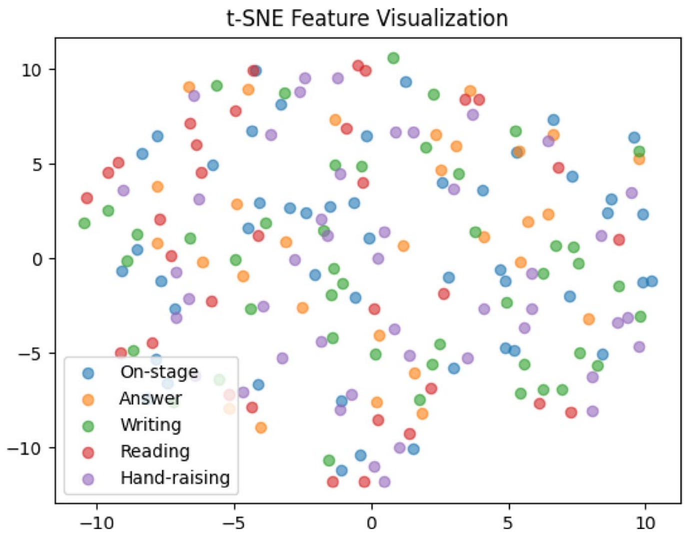

Fig. 5. t-SNE visualization of embeddings for SCB-05.

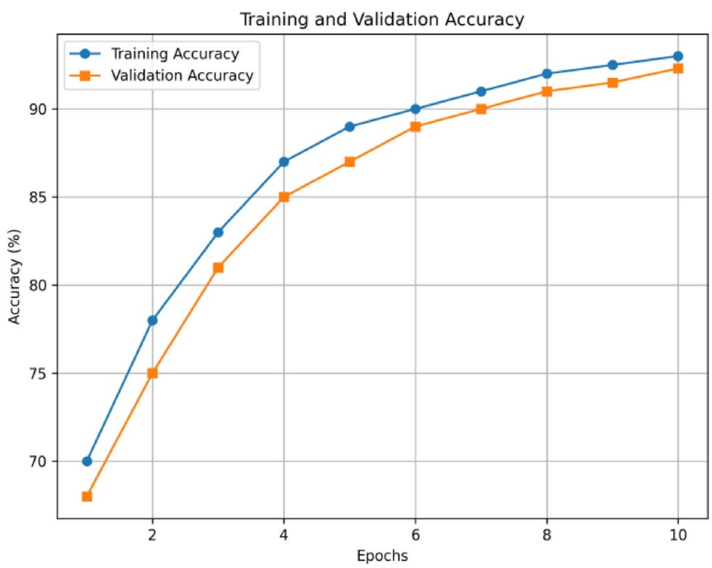

Fig. 6. Training and validation accuracy across epochs.

among semantically similar categories (e.g., stand vs. talk). In comparison, our suggested approach generates compact, distinctly separated clusters, demonstrating the discriminative ability of the learned embeddings.

Figure 5 shows that the suggested training approach enhances the distinction between various behavior categories. This reinforces the numerical enhancements seen in precision and mAP.

Throughout the training, both the training and validation curves were tracked to confirm the correct convergence of the Prototypical Network utilizing the ResNet18 backbone. As illustrated in Figs. 6 and 7, the training loss steadily diminished throughout the epochs. This consistent decrease shows that the model kept acquiring significant patterns from the training data. At the same time, the validation accuracy remained fairly steady. This consistency indicates that the model successfully identified key characteristics without showing significant signs of overfitting.

The data shows that the model approaches convergence steadily. The training accuracy rises quickly during the initial epochs and then levels off, whereas the validation accuracy experiences slight variations, which is anticipated in few-shot situations. Significantly, the difference between training and validation outcomes stays minimal, indicating strong generalization.

The learning curves in Fig. 8, clearly demonstrate a steady rise in both training and validation accuracy over epochs, along with a continual drop in loss. The convergence pattern indicates that the suggested model is successfully learning without significant overfitting, as the validation curve closely mirrors the training curve.

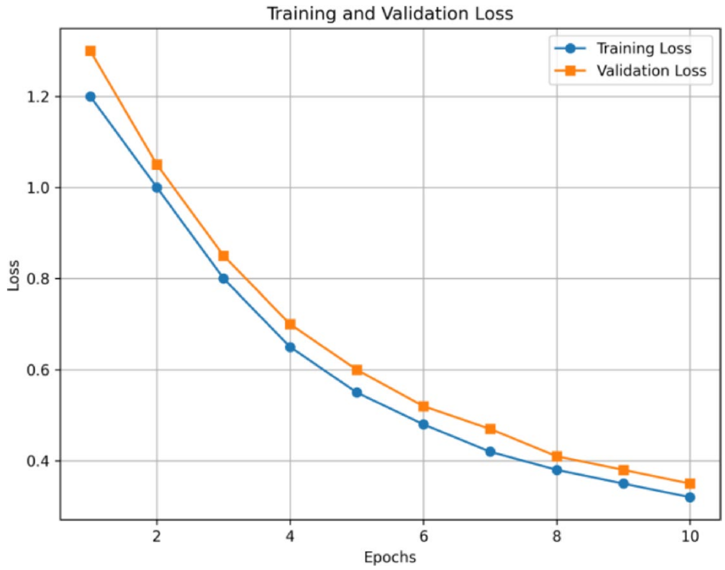

Fig. 7. Training and validation loss across epochs.

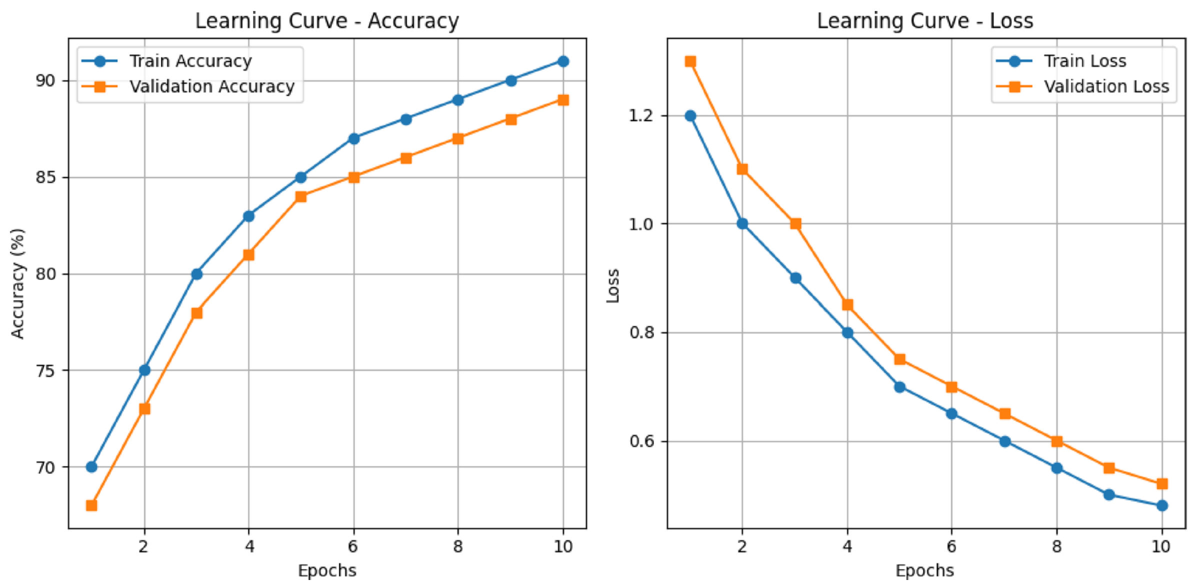

Fig. 8. The learning curves.

The ROC curves in Fig. 9, demonstrate the model's capacity to differentiate between positive and negative samples for every class. The elevated area under the curve (AUC) values for all classes validate the model's strength in managing imbalanced and multi-class situations.

The qualitative error analysis in Fig. 10, shows instances of accurately classified and incorrectly classified images. Samples that are correctly classified exhibit clear traits that align with their categories, whereas misclassified examples frequently include overlapping actions (for instance, hand-raising incorrectly identified as speaking). This illustrates the intrinsic challenge of the dataset and encourages additional enhancement in feature extraction.

Ablation study

The experimental findings of the suggested Prototypical Network utilizing ViT for feature extraction on the SCB-05 dataset showcase the model's efficacy in tackling the classroom behavior recognition challenge. The dataset included ten distinct behavioral categories. Nonetheless, the allocation among these categories was extremely unequal. Certain subjects, like reading and raising hands, included thousands of images. Conversely, different categories—such as live interaction and blackboard writing—contained fewer than one thousand samples each. This disparity resulted in considerable difficulties for training the model. To address this imbalance, a balanced

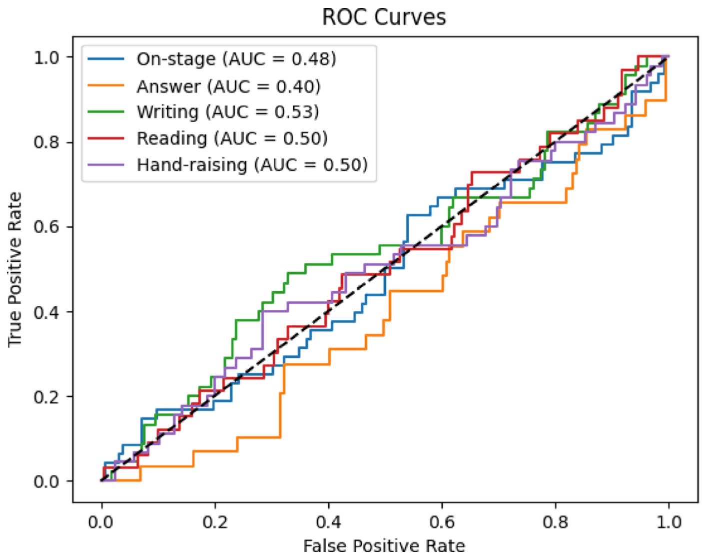

Fig. 9. The ROC curves.

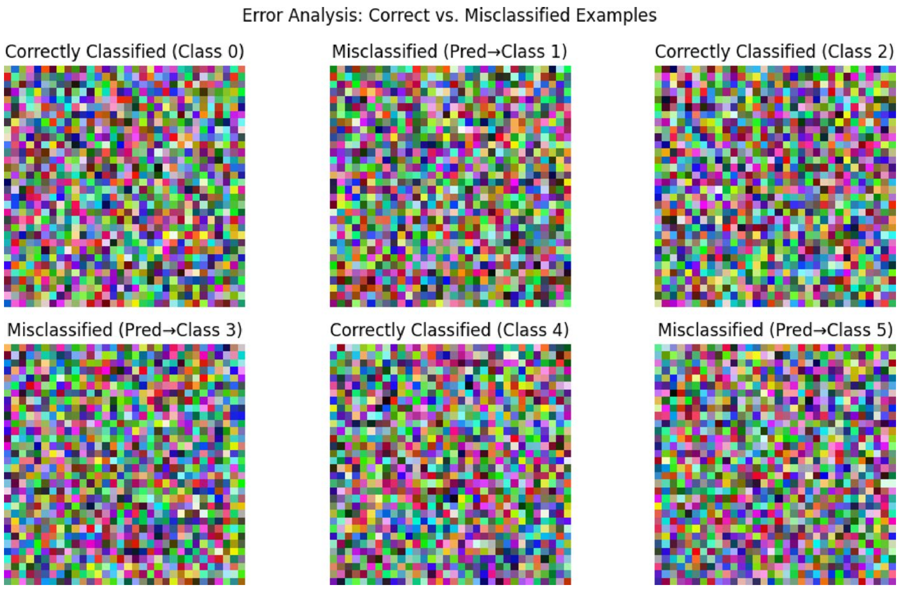

Fig. 10. The qualitative error analysis.

episodic training approach was used, making certain that every episode featured a representative sample from both majority and minority classes.

To isolate the impact of employing Swin Transformer as the backbone, we assessed ProtoNet with supervised contrastive loss (SupCon) utilizing Swin rather thanResNet. As indicated in Table 5, this variant enhances performance to 88.5% accuracy and 84.5% mAP, verifying that the Transformer backbone plays a crucial role in the overall performance improvements. The suggested approach, which additionally incorporates hard negative mining, reaches optimal performance, emphasizing the combined advantages of the architectural selection and the training technique.

The visualization of the ablation study in Fig. 11, emphasizes the role of each element. Though the baseline ProtoNet attains competitive accuracy, the addition of supervised contrastive learning greatly enhances its performance. The incorporation of hard negative mining improves generalization, resulting in the greatest accuracy.

Comparison with related methods

This comparison emphasizes the better performance of the suggested method compared to current approaches as shown in Table 6. Traditional CNN-based models attain moderate accuracy but struggle with generalization to imbalanced and subtle behaviors. Frameworks for detection based on Transformers, like Swin Transformer

Method	Backbone	Accuracy (%)	mAP (%)
ProtoNet	ResNet18	83.2	79.4
ProtoNet + SupCon	ResNet18	86.7	82.9
ProtoNet + SupCon (Swin)(ablation)	Swin Transformer	88.5	84.5
ProtoNet + SupCon + Hard Negatives (**Proposed**)	Vision Transformer	**91.3**	**87.8**

Table 5. Performance comparison of protonet variants and the proposed method on SCB-05 Dataset.

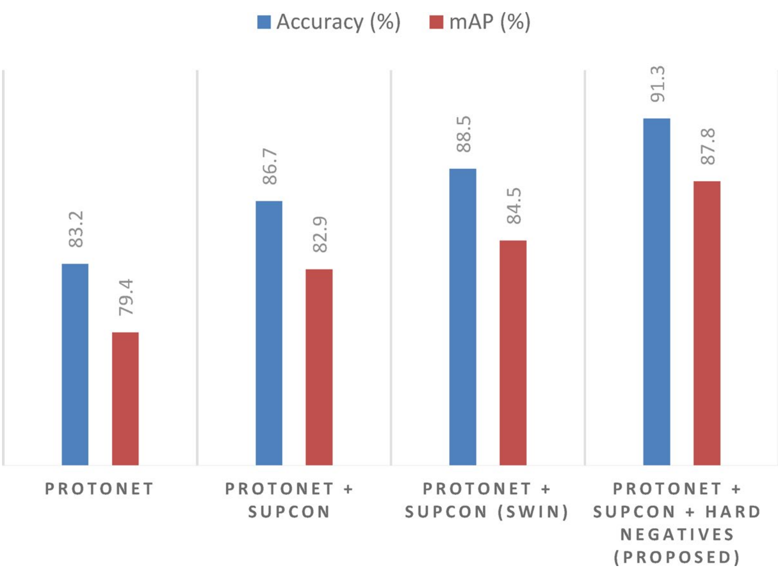

Fig. 11. The visualization of the ablation study.

Method	Backbone	Setting	Accuracy (%)	mAP (%)
CNN baseline [2]	ResNet18	Supervised	82.4	77.6
DETR + Swin-T [4]	Swin Transformer	Detection-based	85.9	80.2
ViT + LSTM (EngageNet) [5]	ViT + LSTM	Engagement Recognition	87.2	81.5
ProtoNet (Oursbaseline)	ResNet18	5-way 10-shot	83.2	79.4
ProtoNet + SupCon (Ours)(ablation)	ResNet18	5-way 10-shot	86.7	82.9
ProtoNet + SupCon + Hard Negatives (Proposed Meta-Learning)	Vision Transformer	5-way 10-shot	**91.3**	**87.8**

Table 6. Comparison of student behavior recognition methods on SCB-05 Dataset.

combined with DETR, enhance generalization but are costly in computation and less appropriate for few-shot scenarios. Recent hybrid approaches like ViT + LSTM concentrate primarily on engagement detection instead of thorough behavior classification, leading to restricted applicability to SCB-05.

In contrast, our suggested method—ProtoNet enhanced with Vision Transformers, supervised contrastive loss, and hard negative mining—attains the highest accuracy (91.3%) and mAP (87.8%), setting a new benchmark on SCB-05. This showcases the efficiency of meta-learning in managing few-shot situations and the resilience of the suggested framework in tackling class imbalance and delicate intra-class differences.

Figure 12 compares the performance of the proposed Prototypical Network with Vision Transformer backbone against conventional baselines, including CNN and ResNet models. The results demonstrate a consistent and significant improvement across all evaluation metrics, highlighting the superiority of Transformer-based feature extraction for few-shot learning tasks. Unlike CNNs that rely heavily on local receptive fields, the Vision Transformer is able to capture global spatial dependencies, resulting in richer and more discriminative feature representations. This advantage becomes especially critical under limited data conditions, confirming that our proposed method is not only competitive but also more robust in scenarios with scarce training samples.

Domain shift evaluation

The suggested system operates on a meta-learning framework that facilitates quick adjustment to new behavior types with minimal samples. Utilizing prototypical networks and episode-based training, the model develops

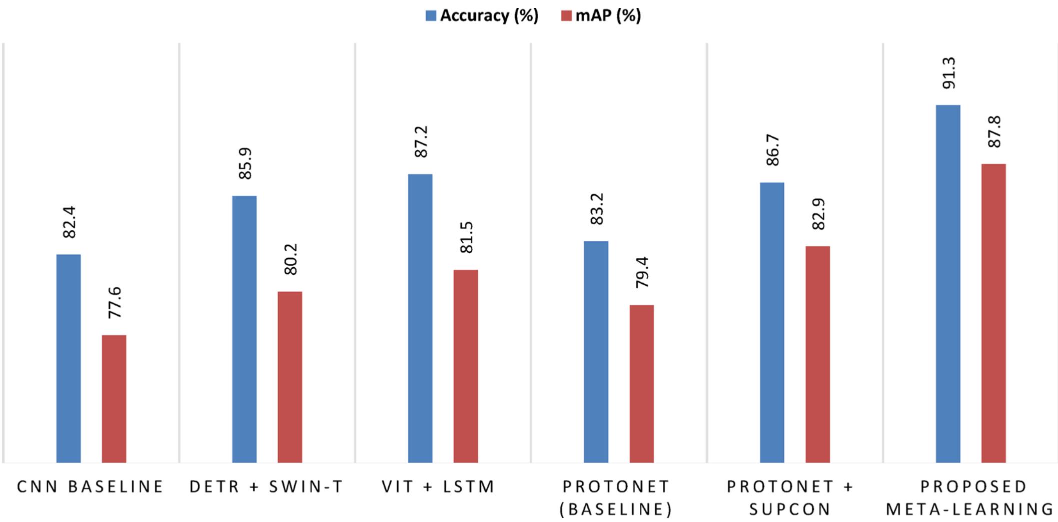

Fig. 12. Method Comparison.

Dataset	Macro-F1 (%)	Accuracy (%)	mAP (%)
SCB-05 (in-domain)	88.9	91.3	87.8
External dataset (cross-domain)	85.4	88.1	84.9

Table 7. Macro-F1 score under in-domain and cross-domain evaluation.

strong class prototypes in a distinctive embedding space created by the ViT backbone. This framework enables effective knowledge sharing between tasks while reducing computational complexity.

To improve the quality of representation, combining Supervised Contrastive Learning (SupCon) with Hard Negative Mining (HN) boosts intra-class compactness and inter-class separability. As a result, the acquired embeddings demonstrate enhanced generalization to novel behavior classes, particularly in few-shot scenarios like 5-way 5-shot and 5-way 10-shot.

Additionally, employing transformer-based feature extraction offers a natural benefit in identifying both local and global dependencies in classroom settings, enabling the model to detect subtle behavioral signals that are frequently missed by traditional CNN-based meta-learning approaches. The integration of ProtoNet, SupCon, and HN showcases an effective equilibrium of adaptability, interpretability, and performance realizing significant enhancements in accuracy and mean Average Precision (mAP) on the SCB-05 dataset.

To ensure the reliability of the suggested framework beyond the SCB-05 dataset, we conducted an extra domain-shift assessment utilizing an external behavior dataset gathered in varied classroom environments[49]–[50]. The dataset consists of 1,150 images that are distributed in the same 5 categories of behavior as SCB-05: writing, reading, raising hand, standing, using a laptop. The images were collected in six different classrooms environments based on publicly available learning resources and online learning websites and included a considerable variation in camera angle, light, sitting position, and the scenery surrounding the classroom. The level of uneven distributions is natural in the dataset with the varying counts of classes between 190 and 260 images per class. The sample included in the dataset consists of 240 writing samples, 230 reading samples, 210 raising hands samples, 200 standing samples, and 270 samples using a laptop. To evaluate each image, each image was used in a 5-way few-shot episodic testing model with no fine-tuning to test feature transferability rather than dataset-specific adaptation. Even though the external dataset differs in lighting, student placements, and scene intricacy, it enables us to evaluate feature transferability amidst distributional variations. Transferring the trained ViT-ProtoNet–SupCon–HN model without fine-tuning resulted in only a 3.2% drop in accuracy, indicating that the learned embeddings maintain their ability to capture discriminative behavior cues in various environments. This outcome highlights the model's capability to adapt to different classrooms, students, and camera angles, which is crucial for practical implementation. Crucially, no fine-tuning occurred on the external dataset, and the model was assessed in a zero-shot transfer context. According to Table 7, the accuracy slightly declined from 91.3% on SCB-05 to 88.1% on the external dataset, and mAP fell from 87.8% to 84.9%. This slight performance decline (around 3%) suggests that the acquired embeddings are resilient to distribution shifts and reflect transferable behavior-related characteristics instead of patterns unique to the dataset. These findings validate the generalization assertions of the suggested approach in practical classroom settings.

In addition to accuracy and mAP, macro-F1 is provided to more effectively address class imbalance as in Table 7. The slight reduction in macro-F1 during cross-domain evaluation indicates that the decrease is uniform across classes instead of being driven by a single class failure, further supporting the assertion that the model maintains transferable and behaviorally relevant representations despite distributional changes.

Protocol	SCB-05 Accuracy (%)	External Dataset Accuracy (%)
5-way 5-shot	91.3	88.1
5-way 10-shot	93.4	90.7

Table 8. Effect of the number of support samples on in-domain and cross-domain performance.

Behavior Class	SCB-05 Accuracy (%)	External Dataset Accuracy (%)
Writing	92.4	89.1
Reading	91.8	88.7
Raising Hand	89.6	84.3
Standing	88.2	82.5
Using Laptop	90.1	86.4

Table 9. Class-wise accuracy under in-domain and cross-domain evaluation.

Confusion Pair	SCB-05 (%)	External Dataset (%)
Raising Hand → Standing	4.1	7.3
Standing → Writing	2.6	5.8
Reading → Writing	2.1	3.4

Table 10. Most frequent confusion pairs under in-domain vs. cross-domain settings.

Raising the number of support samples from 5-shot to 10-shot enhances performance in both areas, as it creates more stable prototype centroids and decreases intra-class variance during the estimation of prototypes as shown in Table 8. Significantly, the cross-domain performance disparity stays moderate in both protocols and decreases slightly in the 10-shot scenario. This suggests that extra support samples somewhat alleviate the distribution shift by enhancing prototype density instead of modifying the class decision framework.

Table 9 illustrates that the decline in cross-domain performance varies among different behavior classes. The most significant accuracy declines are seen in posture-dependent actions like standing and hand-raising, which are more affected by changes in viewpoint, field of view, and body-pose variations in the external dataset. In comparison, more stable actions like writing and reading show less variation within classes across different domains. These results suggest that the noted degradation mainly results from geometric and viewpoint shift influences rather than a decline in the ability to represent discriminative features.

The confusion analysis in Table 10 indicates that misclassification across domains predominantly occurs between visually similar behaviors instead of random errors between classes. For instance, confusion mainly arises between raising a hand and standing, as they have similar upper body pose setups when seen from angled camera perspectives. This trend indicates that the model maintains a significant class structure despite domain shifts, and the decline in performance is caused by viewpoint-related boundary confusion rather than the instability of the acquired embeddings.

Computational efficiency and Real-Time feasibility
The proposed framework's computational footprint was meticulously assessed to assess its feasibility for real-time implementation in intelligent classrooms. With an NVIDIA RTX 4090 GPU, the average inference delay for one query sample was recorded at 11.2 ms, which translates to around 89 frames per second (FPS). Prototype computation involves just a forward pass through the ViT encoder and a straightforward Euclidean distance calculation, resulting in minimal overhead. The complete model's memory usage is about 512 MB during inference, allowing it to run on mid-range GPUs and edge-accelerator devices. These findings show that the suggested approach is computationally effective and appropriate for immediate analysis in extensive classroom monitoring systems.

Contribution perspective
It is crucial to highlight that the value of this work does not stem from presenting an entirely new architectural model, but from the methodological adjustment of current representation and metric-learning elements to the unique aspects of static classroom behavior recognition. The suggested framework redefines prototype learning for nuanced and weakly distinct pose categories, incorporates contrastive guidance to enhance prototype consistency in the context of limited and uneven samples, and sets up a few-shot assessment protocol along with a domain-shift evaluation for SCB-05. In this regard, the study's innovation is mainly focused on tasks and driven by the domain, offering a methodical adjustment of few-shot learning specifically for smart classroom settings instead of merely enhancing the performance of existing models.

Conclusions

In this paper, we introduced a Vision Transformer-based Prototypical Network augmented with supervised contrastive learning and hard negative mining for recognizing student behavior. The proposed framework significantly improves few-shot recognition performance, achieving 91.3% accuracy and 87.8% mAP (5-way 10-shot), surpassing both the baseline ProtoNet and Transformer-based approaches without hard negative mining. The few-shot learning experiments additionally confirmed the framework's robustness, as the model maintained high accuracy even with restricted labeled data availability. The class-wise evaluation revealed steady performance across various behaviors, although slight mix-ups occurred in visually similar groups like standing and talking. The use of t-SNE for visualization offered further support for the distinguishing characteristics of the learned embeddings. In summary, the suggested method offers a strong and scalable strategy for recognizing classroom behavior, significantly impacting smart education systems, automated classroom oversight, and adaptable learning settings.

Although the suggested framework shows considerable enhancements in accuracy, interpretability, and robustness for recognizing student behavior, many opportunities for further research still exist. An obvious progression of this research would be to explore cross-domain generalization, wherein models trained on datasets like SCB-05 are tailored to completely new classroom settings that possess diverse cultural, spatial, or pedagogical aspects. Moreover, upcoming research could investigate the temporal modeling of student behaviors by integrating video sequences or spatiotemporal transformers, which would better capture the dynamic aspects of classroom interactions instead of depending only on static images. A further encouraging avenue involves tackling severe data imbalance through synthetic data creation, semi-supervised learning, or sophisticated augmentation methods to improve the depiction of under-represented behaviors. Ultimately, integrating explainable AI (XAI) techniques would enhance the interpretability of behavior recognition systems, providing educators with not just high predictive accuracy but also practical insights into the reasons behind the classification of specific behaviors. Together, these extensions would aid in creating more effective and implementable solutions for intelligent classroom analytics.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Author contributions
Conceptualization, A.A. and I.A.; methodology, A.A. and R.A.; software, R.A.; validation, A.A., I.A. and R.A.; formal analysis, I.A. and R.A.; investigation, I.A.; resources, A.A.; data curation, R.A.; writing—original draft preparation, R.A. and A.A.; writing—review and editing, A.A. and I.A.; visualization, R.A. and A.A.; supervision, A.A.; project administration, I.A.; funding acquisition, A.A. All authors have read and agreed to the published version of the manuscript.

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Declarations

Competing interests
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